

TIER 1 — CORE FUNDAMENTALS & LOCATORS

ASSIGNMENT 1 — The Multi-Locator Challenge

1. Experiment Title

Multi-Locator Web Element Identification Using Selenium

2. Problem Statement

Navigate to the SauceDemo login page and interact with the username field using By.ID, the password field using By.NAME, and the login button using By.XPATH. Verify successful login by checking that the resulting URL contains /inventory.html.

3. Objective

- To understand different Selenium locator strategies.
- To use By.ID, By.NAME, and By.XPATH.
- To automate a login process.
- To validate the resulting URL using an assertion.

4. Tools, Software and Concepts Used

Tools & Software:

- Python
- Selenium WebDriver
- Google Chrome
- VS Code / PyCharm

Concepts:

- Selenium WebDriver
- By.ID
- By.NAME
- By.XPATH
- send_keys()
- click()

- Assertions
- URL Validation

5. Source Code / Implementation Steps

```
from selenium import webdriver

from selenium.webdriver.common.by import By
import time

def test_multi_locator_login():

    # Launch browser

    driver = webdriver.Chrome()

    driver.maximize_window()

    # Open SauceDemo

    driver.get("https://www.saucedemo.com/")

    # Username using By.ID

    username = driver.find_element(By.ID, "user-name")

    username.send_keys("standard_user")

    # Password using By.NAME

    password = driver.find_element(By.NAME, "password")

    password.send_keys("secret_sauce")

    # Login button using By.XPATH

    login_button = driver.find_element(

        By.XPATH, "//input[@type='submit']"

    )

    login_button.click()

    # Validate resulting URL

    assert "/inventory.html" in driver.current_url
```

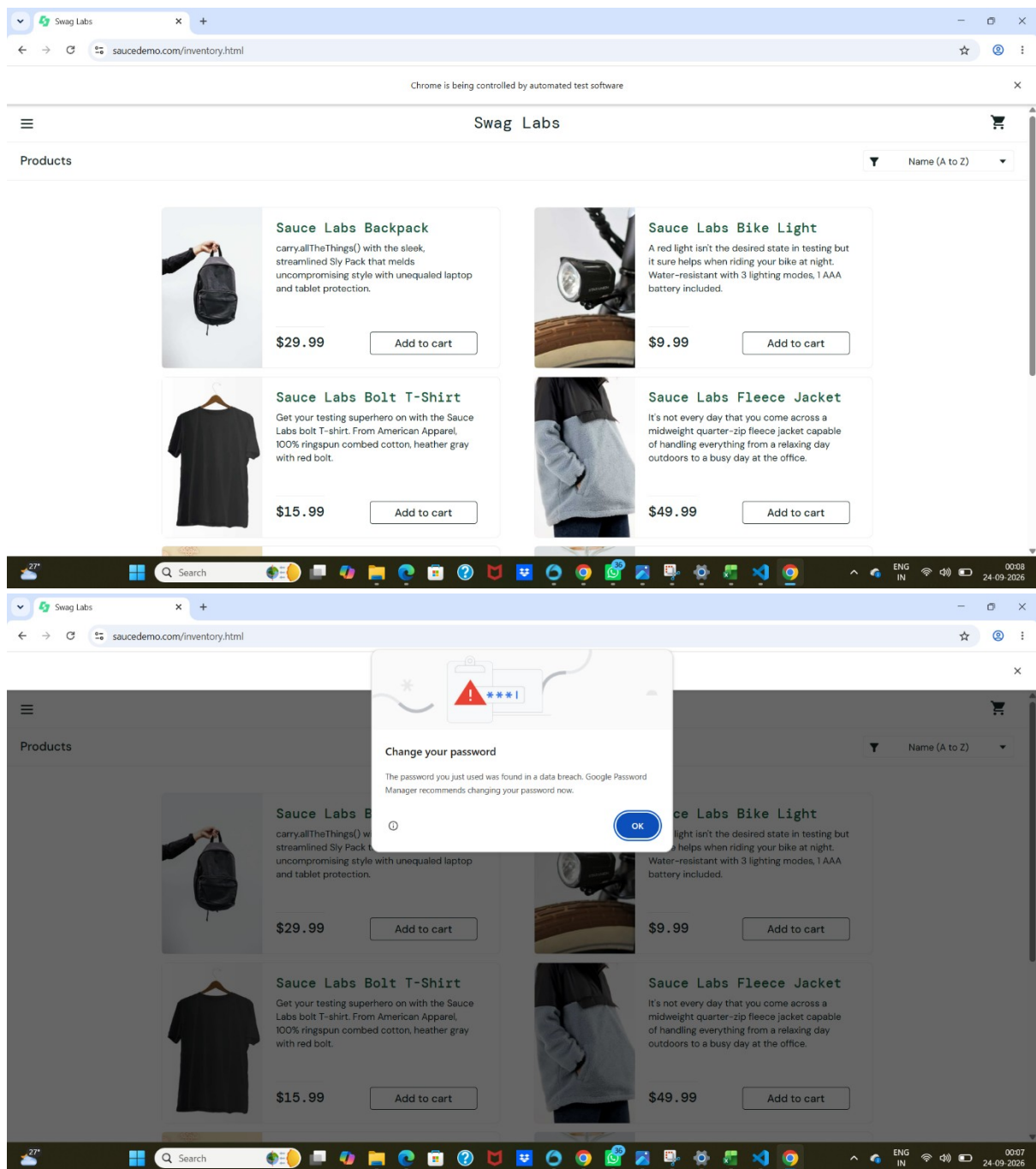
```
# Keep browser open for screenshot
```

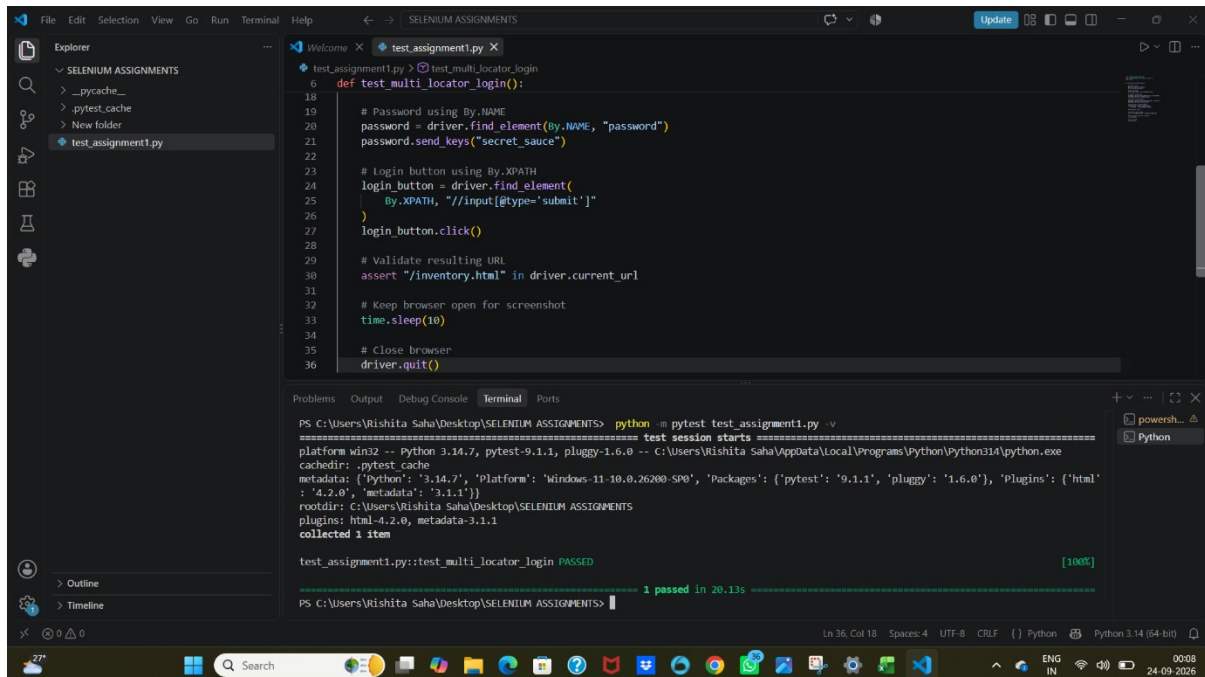
```
time.sleep(10)
```

```
# Close browser
```

```
driver.quit()
```

6. Output with Clearly Labelled Screenshots





Show the terminal displaying successful execution.

7. Result

The SauceDemo login page was successfully automated using three different Selenium locator strategies, and the resulting URL was successfully validated.

8. Observation

- By.ID was used for the username field.
- By.NAME was used for the password field.
- By.XPATH was used for the login button.
- The URL assertion confirmed successful login.

9. Conclusion

This experiment demonstrated the use of multiple Selenium locator strategies for identifying and interacting with web elements.